# *Chapter 3*
# *Vector Spaces*

## 3.4 Basis and Dimension

# Span and Linear Independence

**Definition:**

The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent (线性无关)** if

$$\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n = 0$$

implies that all the scalers $\alpha_1, \alpha_2, \ldots, \alpha_n$ **must equal to** 0.

**Definition:**

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The set of all possible linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is called the **span** of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, denoted by $span(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n)$.

$$span(\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n) = \{\alpha_1\boldsymbol{v}_1 + \alpha_2\boldsymbol{v}_2 + \cdots + \alpha_n\boldsymbol{v}_n : \alpha_i \in R\}$$

**Theorem 3.3.2:** A vector $\mathbf{v} \in$ Span($\mathbf{v}_1$, …, $\mathbf{v}_n$) can be written *uniquely* as a linear combination of $\mathbf{v}_1$, …, $\mathbf{v}_n$ if and only if $\mathbf{v}_1$, …, $\mathbf{v}_n$ are linearly independent.

# Basis

**Definition**

The vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ form a **basis** for a vector space $V$ if and only if

1. $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are **linearly independent**.
2. $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ **span** $V$.

- If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ form a basis for $V$, then:
  for any $\boldsymbol{b} \in V$, there exists unique $\alpha_1, \alpha_2, \ldots, \alpha_n$, such that

$$\boldsymbol{b} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2, + \cdots + \alpha_n \mathbf{v}_n$$

# Example 1: some basis for $\mathbb{R}^3$

- The vectors $[1, 0, 0]^T$, $[0, 1, 0]^T$, $[0, 0, 1]^T$ form a **basis** for $\mathbb{R}^3$ (called the standard basis) because:

1. They **span** $\mathbb{R}^3$. That is, we can write an arbitrary vector $[x, y, z]^T$ as a **linear combination** of the three vectors:
   For any $[x, y, z]^T \in \mathbb{R}^3$, $[x, y, z]^T = x[1,0,0]^T + y[0,1,0]^T + z[0,0,1]^T$
2. They are **linearly independent**.
   If $p[1,0,0]^T + q[0,1,0]^T + r[0,0,1]^T = [0,0,0]^T$ then $p = q = r = 0$ is the unique solution.

# Example 1 (cont.)

The standard basis for $\mathbf{R}^3$ is $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$; however, there are many bases that we could choose for $\mathbf{R}^3$. For example,

$$\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}0\\1\\1\end{pmatrix}, \begin{pmatrix}2\\0\\1\end{pmatrix}\right\} \quad \text{and} \quad \left\{\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}1\\1\\0\end{pmatrix}, \begin{pmatrix}1\\0\\1\end{pmatrix}\right\}$$

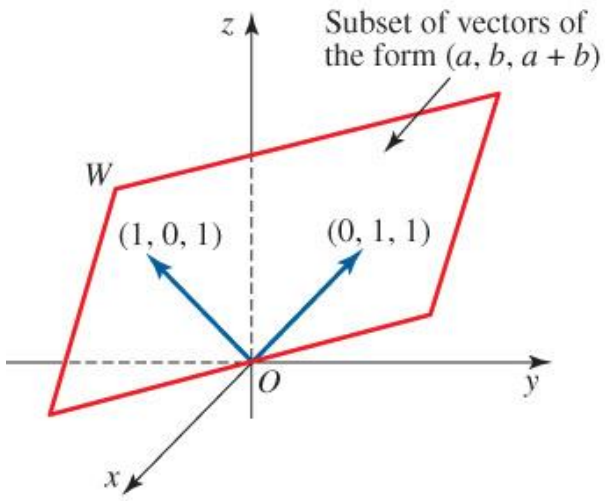
are both bases for $\mathbf{R}^3$. We will see shortly that

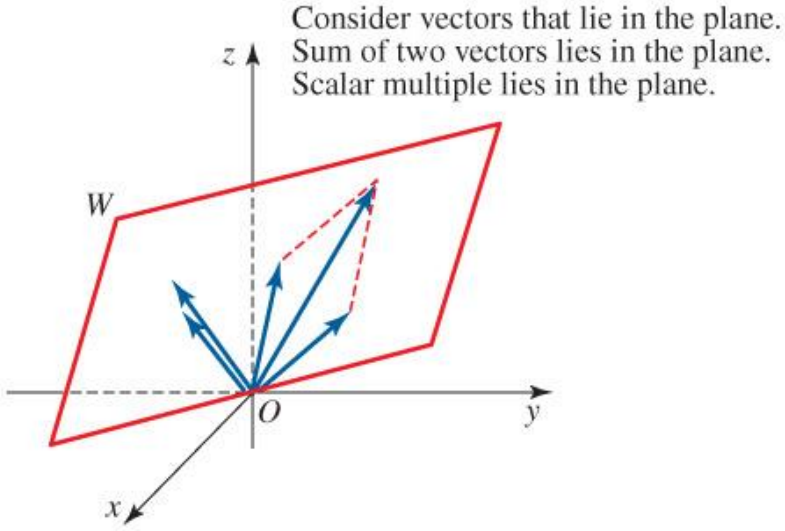
- any basis for $\mathbf{R}^3$ must have exactly *three* elements.
- <= 2 vectors cannot span $\mathbf{R}^3$
- >=4 vectors in $\mathbf{R}^3$ cannot be linearly independent.

# Example 2: find basis for a subspace

Let $W$ be the subspace consisting of vectors of the form $\mathbf{u} = [a, b, a+b]^T$. Find a **basis** for $W$.

- **Span:** $\mathbf{u} = [a, b, a+b]^T = [a, 0, a]^T + [0, b, b]^T = a[1,0,1]^T + b[0,1,1]^T$, so the vectors $[1,0,1]^T$ and $[0,1,1]^T$ span $W$.
- **Linear independence:** $p[1,0,1]^T + q[0,1,1]^T = [0,0,0]^T$ leads to $p = 0$ and $q = 0$, the two vectors are thus linearly independent.
- **Conclusion:** The set $\{[1,0,1]^T, [0,1,1]^T\}$ is therefore a basis for $W$.



# Example 3: find basis for a subspace

Consider the subset *V* of $\mathbb{R}^3$ of vectors of the form $[a, 2a, 3a]^T$. Show that *V* is a subspace of $\mathbb{R}^3$ and find a basis.

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**Solution**

(subspace)

- Let $\mathbf{u} = [a, 2a, 3a]^T$ and $\mathbf{v} = [b, 2b, 3b]^T$ be vectors in *V* and *k* be a scalar.

1) $\mathbf{u}+\mathbf{v} = [a+b, 2(a+b), 3(a+b)]^T \quad \mathbf{u}+\mathbf{v} \in V$

2) $k\mathbf{u} = k[a, 2a, 3a]^T = [ka, 2ka, 3ka]^T \quad \therefore k\mathbf{u} \in V$

*V* is closed under addition and scalar multiplication.

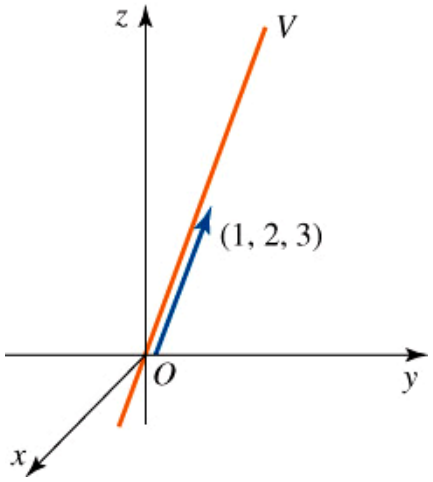
*V* is non-empty because $[0, 0, 0]^T$ is in *V*

Therefore *V* is a subspace of $\mathbb{R}^3$.

(basis)

- $\mathbf{u} = [a, 2a, 3a]^T = a[1, 2, 3]^T$

$\Rightarrow \{[1, 2, 3]^T\}$ is a basis for *V*.



# Standard Basis of $\mathbb{R}^n$

The set $\{e_1 = [1, 0, \dots, 0]^T, e_2 = [0, 1, \dots, 0]^T, \dots, e_n = [0, \dots, 1]^T\}$ of *n* vectors is the ***standard basis*** for $\mathbb{R}^n$.

$\mathbb{R}^2$: two-dimensional space
$\mathbb{R}^3$: three-dimensional space

- Standard basis for $\mathbb{R}^3$: $\{e_1, e_2, e_3\}$.
- Standard basis for $\mathbb{R}^{2\times 2}$:
  $\{E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}\}$.
- Standard basis for $P_4$: $\{1, x, x^2, x^3\}$.

# Dimension

**Definition**

- Let $V$ be a vector space. If $V$ has a basis consisting of $n$ vectors, we say that $V$ has *dimension* $n$ and write $\dim V = \mathrm{n}$
- The subspace $\{\mathbf{0}\}$ of $V$ is said to have dimension 0.
- $V$ is *finite dimensional* if there is a finite set of vectors that spans $V$; otherwise $V$ is *infinite dimensional*.

Examples:

- $\dim \mathbb{R}^n = n$
- $\dim \mathbb{R}^{m\times n} = m \times n$
- $\dim P_n = n$

# Example: Dimension of Null Space

$\boldsymbol{A} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix}$. Find the basis of $N(A)$ and its dimension.

**Solution**

Use Goss-Jordan reduction to solve $A\mathbf{x} = 0$, we obtain

$$[\boldsymbol{A}|\mathbf{0}] = \left[\begin{array}{cccc|c} 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 \end{array}\right]$$

How to determine leading variables and free variables?

The reduced echelon form involves two free variables, $x_3, x_4$:

$$x_1 = x_3 - x_4$$
$$x_2 = -2x_3 + x_4$$

Write leading variables in terms of free variables

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_3 - x_4 \\ -2x_3 + x_4 \\ x_3 + 0x_4 \\ 0x_3 + x_4 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Write all the variables in a column vector, and replace the leading variable by free variables

Then, $N(A) = \{\mathbf{x}: \mathbf{x} = \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}\}$

**Dim(N(A))=2**

**Basis vectors are** $\begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

# Example: nullspace, dimension, basis

Consider the following system of homogeneous linear equations.

$$\begin{aligned} x_1 - x_2 + x_3 + 2x_4 &= 0 \\ x_1 - 3x_3 + 2x_4 &= 0 \\ 2x_1 - x_2 - 2x_3 + 4x_4 &= 0 \end{aligned}$$

1. **Find the null space of the linear equations.**
2. **Find the dimension of the null space**
3. **Find the basis of the null space**

# Example: nullspace, dimension, basis

Consider the following system of homogeneous linear equations.

$$\begin{aligned} x_1 - x_2 + x_3 + 2x_4 &= 0 \\ x_1 - 3x_3 + 2x_4 &= 0 \\ 2x_1 - x_2 - 2x_3 + 4x_4 &= 0 \end{aligned}$$

1. **Find the null space of the linear equations.**
2. **Find the dimension of the null space**
3. **Find the basis of the null space**

**Sol**

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 0 \\ 1 & 0 & -3 & 2 & 0 \\ 2 & -1 & -2 & 4 & 0 \end{array}\right] \xrightarrow[\text{R3-2R1}]{\text{R2-R1}} \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 0 \\ 0 & 1 & -4 & 0 & 0 \\ 0 & 1 & -4 & 0 & 0 \end{array}\right] \xrightarrow{\text{R3-R2}} \left[\begin{array}{cccc|c} 1 & -1 & 1 & 2 & 0 \\ 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\text{R1+R2}}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 2 & 0 \\ 0 & 1 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 3x_3 - 2x_4 \\ 4x_3 + 0x_4 \\ x_3 + 0x_4 \\ 0x_3 + x_4 \end{bmatrix} = x_3 \begin{bmatrix} 3 \\ 4 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 3 \\ 4 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} x_1 - 3x_3 + 2x_4 &= 0 \\ x_2 - 4x_3 &= 0 \end{aligned}$$

$$N(A) = \left\{ \boldsymbol{u} \mid \boldsymbol{u} = \alpha \begin{bmatrix} 3 \\ 4 \\ 1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

**Dim(N(A))=2**

**Basis vectors are**

$$\begin{bmatrix} 3 \\ 4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

# Example

- Find a basis for N(A) where

$$A = \begin{bmatrix} 4 & 9 & 1 & -3 \\ 0 & -2 & -2 & -2 \\ 1 & 1 & -1 & -2 \end{bmatrix}$$

## Solution

- The reduced echelon form of S is:

$$U = \begin{bmatrix} 1 & 0 & -2 & -3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

- $N(A) = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \end{bmatrix} x_4 \middle| x_3, x_4 \in \mathbb{R} \right\} = span\left( \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right)$
- The two vectors are linearly independent; therefore, the two vectors form a bas for N(A).

# Question: are there quicker ways?

Are there quick ways to answer the following questions?

- Is $\left\{\begin{bmatrix}1\\2\\0\end{bmatrix}, \begin{bmatrix}0\\1\\-1\end{bmatrix}, \begin{bmatrix}1\\1\\2\end{bmatrix}\right\}$ a basis for $\mathbb{R}^3$?
- Is $\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}0\\1\\2\end{bmatrix}\right\}$ a basis for $\mathbb{R}^3$?

Using definition, to check if a set of vectors is a **basis**, we need to check whether it is a **spanning set** (check whether equations are consistent for all **b**) AND whether it is **linearly independent** (solve homogeneous equation). **Are there quicker ways?**

**Next we will list some theorems to help us answer these questions**

# Theorem 3.4.1

If S = $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set for a vector space $V$ , then any collection of $m$ vectors in $V$ with $m > n$ is linearly dependent.

**Explanation**

- Intuitively: if n vectors already span S, you don't need more vectors.
- Textbook P142.
- Proof Idea: n homogenous equations with m>n unknowns must have non-trivial solutions (because of free variables).

# Important Corollary (3.4.2)

If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ and $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ are both bases for a vector space $V$, then $m = n$.

## Proof

- Since $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is a spanning set for $V$, and set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ is linearly independent, by theorem we have $m \leq n$.
- By the same reasoning, we have $n \leq m$.
- Thus $m = n$.

## Explanation

A vector space can have many basis, but they all have the same number of vectors (which is the dimension of V)

# Theorem 3.4.3: explanation

If $V$ is a vector space with $\dim V = n > 0$, then

1. any set of $n$ linearly independent vectors also spans $V$;
2. any $n$ vectors that span $V$ are also linearly independent.

## Explanation

- How to use this theorem: If you already know the dimension of V is n, then a basis can only consist of n vectors.
  (> n vectors: linearly dependent, not a basis;
  <n vectors: cannot span, not a basis.)
- To check whether n vectors is a basis of an n-dimensional V, we only need to check 1 of the 2 conditions of basis: linearly independent ⇔ span.
- WARNING: only works for n vectors in n-dimensional vector space.

# Theorem 3.4.3: Proof of 1

If $V$ is a vector space with $\dim V = n > 0$, then

1. any set of $n$ linearly independent vectors also spans $V$;
2. any $n$ vectors that span $V$ are also linearly independent.

## Proof of 1

- Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent and $\mathbf{v}$ is any other vector in $V$. By theorem1, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n, \mathbf{v}$ are linearly dependent. Therefore, exist not all zero scalers

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n + c_{n+1}\mathbf{v} = 0$$

- $c_{n+1}$ cannot be zero (otherwise $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ would be linearly dependent)
- Therefore any vector $\mathbf{v}$,

$$\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \cdots + \alpha_n\mathbf{v}_n, \quad \text{where } \alpha_i = -{}^{c_i}/_{c_{n+1}}$$

# Theorem 3.4.3: Proof of 2

If $V$ is a vector space with $\dim V = n > 0$, then

1. any set of $n$ linearly independent vectors also spans $V$ ;
2. any $n$ vectors that span $V$ are also linearly independent.

**Proof of 2**

- Suppose $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ span $V$. If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly dependent. Then one of the vectors , say $\boldsymbol{v}_n$, can be written as a linear combination of the others.
- It follows that $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{n-1}$ will still span $V$.
- We can continue this process until reach a linearly independent set with $k < n$ elements, which contradicts $\dim V = n$.

# Example

- Is $\left\{\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}-2\\1\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}\right\}$ a basis for $\mathbb{R}^3$? Why?

**Solution**

- Since dim $\mathbb{R}^3$=3, we need only to test whether the three vectors are linearly independent.
- Since,

$$\begin{vmatrix}1 & -2 & 1\\2 & 1 & 0\\3 & 0 & 1\end{vmatrix} = 2$$

for a square matrix, det is not 0 => matrix is nonsingular => columns are linearly independent. (theorem 3.3.1)

- Therefore the three vectors form a basis for $\mathbb{R}^3$

# Theorem 3.4.4

- If V is a vector space with dim V = n > 0, then
  a) no set with k < n vectors can span V ;
  b) any subset of k < n linearly independent vectors can be extended to form a basis for V ;
  c) any spanning set of V with m > n vectors can be cut down to form a basis for V .

**Proof idea**

a) The span of k<n vectors have dimension <=k, not n.

b) Keep adding in a vector not already in the span
(the new set is still linearly independent).

c) Keep removing a vector that is already in the span of other vectors
(the new set still spans).

# Example

- Extend $\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}0\\1\\2\end{bmatrix}\right\}$ to form a basis of $\mathbb{R}^3$.
- Hint: can you add in 1 more column vector, so that the matrix

$$X = \begin{bmatrix}1 & 0 & ?\\2 & 1 & ?\\1 & 2 & ?\end{bmatrix}$$

has $\det(X) \neq 0$?

- For example if the third column is $\begin{bmatrix}0\\0\\1\end{bmatrix}$, then $\det(X) = 1 \neq 0$

  (Why? How to calculate the determinant of a triangular matrix?),

  therefore $\left\{\begin{bmatrix}1\\2\\1\end{bmatrix}, \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$ forms a basis of $\mathbb{R}^3$.

# Exercise

State with a brief explanation whether the following statements are true or false.

(a) The set $\{[1,0,0]^T, [0,1,0]^T\}$ is a basis for a two-dimensional subspace of $\mathbb{R}^3$.

(b) The set $\{\ [1,0,0]^T\ \}$ is a basis for a one-dimensional subspace of $\mathbb{R}^3$.

(c) The set of all vectors of the form $[a, 2a, b]^T$ is a subspace spanned by the vectors $[1,2,0]^T$ and $[0,0,1]^T$.

(d) The set of vectors of the form $[a, b, 2a - b]^T$ is a vector in the plane spanned by the vectors $[1,0,2]^T$ and $[0,1,-1]^T$.

(e) The set $\{[1,0,0]^T, [0,1,0]^T, [1,1,0]^T\}$ is a basis for a subspace of $\mathbb{R}^3$.

# Exercise

State with a brief explanation whether the following statements are true or false.

(a) The set $\{[1,0,0]^T, [0,1,0]^T\}$ is a basis for a two-dimensional subspace of $\mathbb{R}^3$. True

(b) The set $\{ [1,0,0]^T \}$ is a basis for a one-dimensional subspace of $\mathbb{R}^3$.

True

(c) The set of all vectors of the form $[a, 2a, b]^T$ is a subspace spanned by the vectors $[1,2,0]^T$ and $[0,0,1]^T$. True

(d) The set of vectors of the form $[a, b, 2a - b]^T$ is a vector in the plane spanned by the vectors $[1,0,2]^T$ and $[0,1,-1]^T$. True

(e) The set $\{[1,0,0]^T, [0,1,0]^T, [1,1,0]^T\}$ is a basis for a subspace of $\mathbb{R}^3$.

False, v3 = v1+v2, linearly dependent, cannot be a basis